NET732

Week8:Secure Network Design

Practical

1. Checking the network interface

I started by checking my Ubuntu network interfaces using:

ip -br link

This helped me identify my active network interface.

2. Enabling VLAN support

I enabled VLAN support using:

sudo modprobe 8021q

This allowed my Ubuntu system to create VLAN interfaces.

3. Creating two VLANs

I created two VLAN interfaces using the following commands:

sudo ip link add link ens33 name ens33.10 type vlan id 10

sudo ip link add link ens33 name ens33.20 type vlan id 20

I assigned IP addresses to both VLANs:

sudo ip addr add 192.168.10.1/24 dev ens33.10

sudo ip addr add 192.168.20.1/24 dev ens33.20

I then activated both interfaces:

sudo ip link set ens33.10 up

sudo ip link set ens33.20 up

4. Checking VLAN configuration

I used the following command to check whether my VLANs were created successfully:

ip -d link show type vlan

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$ ip -d link show type vlan
3: ens33.10@ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff promiscuity 0 allmulti 0 minmtu 0 maxmtu 65535
    vlan protocol 802.1Q id 10 <REORDER_HDR> addrgenmode none numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65
_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
4: ens33.20@ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether 00:0c:29:4a:af:9c brd ff:ff:ff:ff:ff:ff promiscuity 0 allmulti 0 minmtu 0 maxmtu 65535
    vlan protocol 802.1Q id 20 <REORDER_HDR> addrgenmode none numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65
_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$
```

This displayed my VLAN interfaces and their VLAN IDs.

5. Configuring firewall rules

I installed UFW and configured access control rules.

sudo apt install ufw -y

I allowed SSH traffic on VLAN 10 and blocked SSH traffic on VLAN 20:

sudo ufw allow in on ens33.10 to any port 22 proto tcp

sudo ufw deny in on ens33.20 to any port 22 proto tcp

```
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$ sudo apt install ufw -y
ufw is already the newest version (0.36.2-9build1).
ufw set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30          linux-image-unsigned-7.0.0-30-generic    linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$ sudo ufw allow in on ens33.10 to any port 22 proto tcp
Rules updated
Rules updated (v6)
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$ sudo ufw allow in on ens33.20 to any port 22 proto tcp
Rules updated
Rules updated (v6)
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$ sudo ufw status numbered
Status: inactive
maso-prince@maso-prince-VMware-Virtual-Platform:~/CyberOps/Week8$
```

I checked my firewall rules using:

sudo ufw status numbered

6. Saving evidence and logs

I saved my VLAN configuration and firewall rules using:

ip -d link show type vlan > vlan_config.txt

sudo ufw status numbered > firewall_rules.txt

I also recorded my terminal activities in commands.log.

7. Analysis and conclusion

During this practical, I created two VLAN interfaces and configured firewall rules to control network access. I learned how VLANs separate network traffic and how firewalls help control access between network interfaces.

This is important in SOC operations because network segmentation and access controls help protect systems from unauthorised access and limit the spread of security threats.

I successfully completed the VLAN and firewall configuration activities and saved my evidence. My next step is to practise testing communication between segmented networks and improving network security.